

Day 5 — Adult Income Final Submission

Reproducible binary classification project

Task 1 — Reproducible Pipeline

Dataset: 32,561 rows, 14 predictors. Stratified 60/20/20 split with random_state=42. Numeric preprocessing: median imputation + scaling. Categorical preprocessing: most-frequent imputation + one-hot encoding.

Task 2 — Hyperparameter Search

Five-fold StratifiedKFold with F1 scoring. Mean CV F1: Logistic Regression 0.6811; Random Forest 0.6784; Gradient Boosting 0.6792.

Task 3 — Learning Curves

Learning curve generated on train/development data only; figure included in figures/.

Task 4 — Calibration and Threshold

Threshold selected only on development data. Selected threshold = 0.66. Final calibration assessed with Brier score.

Task 5 — Final Evaluation & Save Artifact

Final test metrics: Accuracy 0.8408, Precision 0.6595, Recall 0.7011, F1 0.6796, ROC-AUC 0.9016, PR-AUC 0.7597, Brier 0.1308. Confusion matrix TN=4376, FP=568, FN=469, TP=1100.

Final model: Logistic Regression. The fitted pipeline and selected threshold are stored in adult_income_final_pipeline.joblib. The final test set was not used for selection or threshold tuning.